Class 5: Data Viz with ggplot

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Background

There are lot's of ways to make plots in R. These include so-called "base R" (like the `plot()`) and add on packages like **ggplot2**.

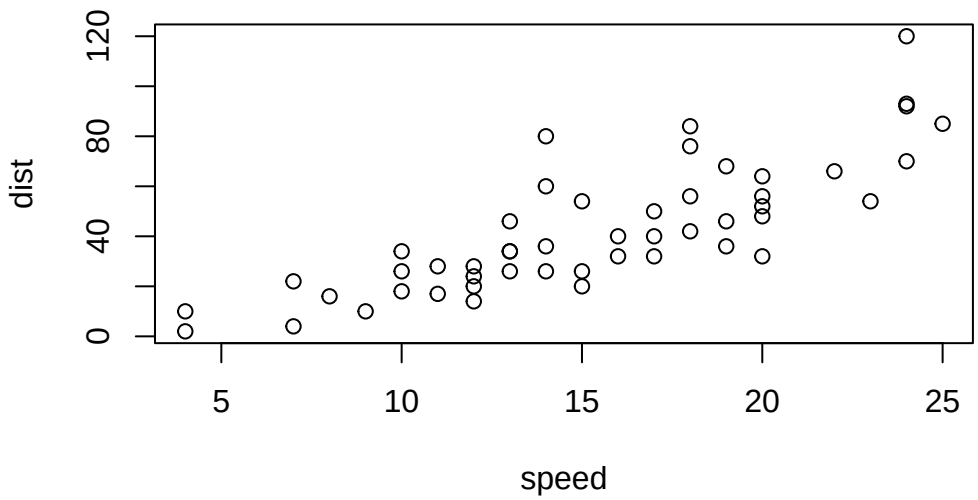
Let's make the same plot with these two graphics systems. We can use the inbuilt `cars` dataset:

```
head( cars )
```

```
  speed dist
1     4    2
2     4   10
3     7    4
4     7   22
5     8   16
6     9   10
```

With "base R" we can simply:

```
plot(cars)
```



Now let's try ggplot. First I need to install the package using `install.packages("ggplot2")`.

> **N.B.** We never run an `installl.packages()` in a code chunk otherwise we will re-install needlessly every time we render our document.

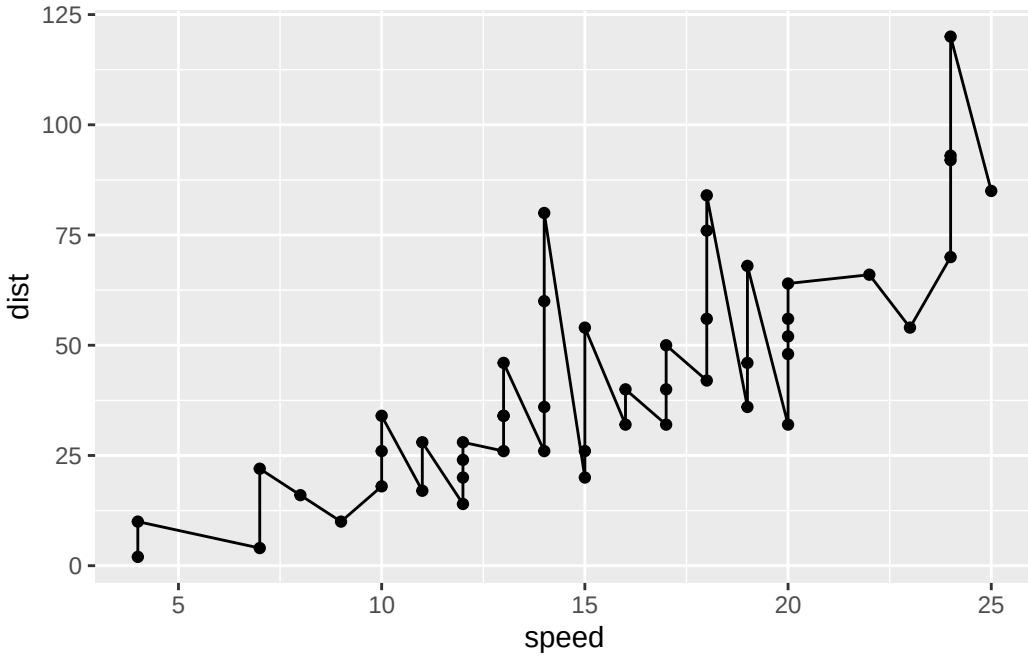
Every time we want to use an add-on package we need to load it up with a call to `library()`

```
library(ggplot2)
ggplot(cars)
```

Every ggplot needs at least 3 things:

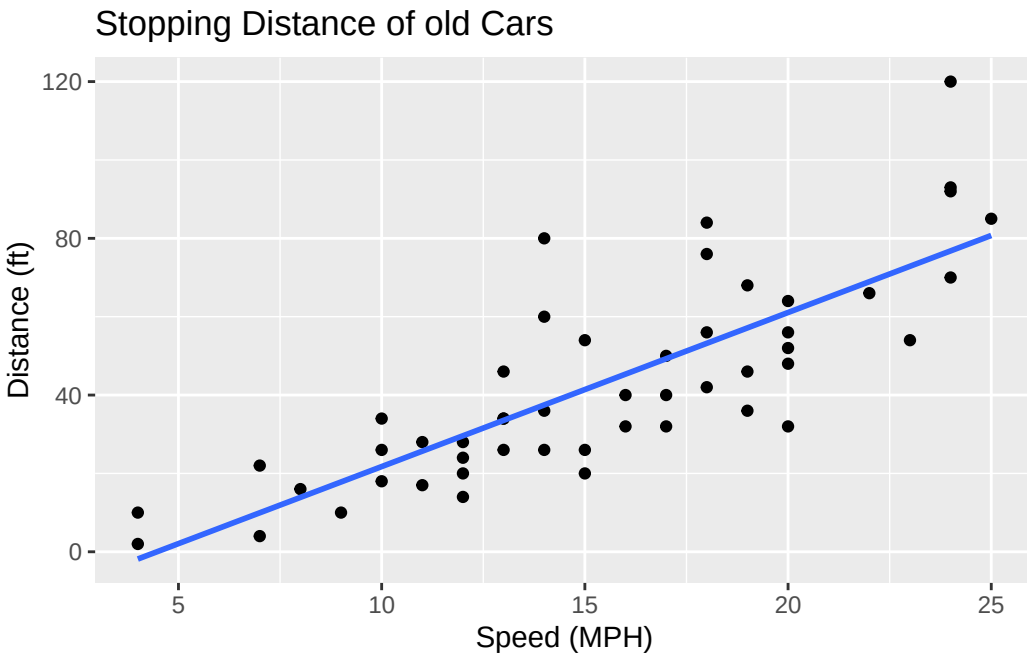
1. The **data** i.e. stuff to plot as a data.frame
2. The **aes** for aesthetics that map the data to the plot
3. The **geom_** or geometry i.e. the plot type such as points, lines etc.

```
ggplot(cars) +
  aes(x=speed, y=dist) +
  geom_point() +
  geom_line()
```




```
ggplot(cars) +
  aes(x=speed, y=dist) +
  geom_point() +
  geom_smooth(method = "lm", se = FALSE) +
  labs(x="Speed (MPH)",
y="Distance (ft)",
title="Stopping Distance of old Cars")
```

```
`geom_smooth()` using formula = 'y ~ x'
```




```
theme_bw()
```

```
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  ..@ colour       : chr "black"
  ..@ linewidth    : num 0.5
  ..@ linetype     : num 1
  ..@ lineend      : chr "butt"
  ..@ linejoin     : chr "round"
  ..@ arrow        : logi FALSE
  ..@ arrow.fill   : chr "black"
  ..@ inherit.blank: logi TRUE
 $ rect                            : <ggplot2::element_rect>
  ..@ fill         : chr "white"
  ..@ colour       : chr "black"
  ..@ linewidth    : num 0.5
  ..@ linetype     : num 1
  ..@ linejoin     : chr "round"
  ..@ inherit.blank: logi TRUE
 $ text                            : <ggplot2::element_text>
  ..@ family       : chr ""
  ..@ face         : chr "plain"
```

```
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 ..@ fontweight   : num NA
 ..@ fontwidth    : num NA
 ..@ colour       : chr "black"
 ..@ size         : num 11
 ..@ hjust        : num 0.5
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 ..@ hjust        : NULL
 ..@ vjust        : NULL
 ..@ angle        : NULL
 ..@ lineheight   : NULL
 ..@ margin       : NULL
 ..@ debug        : NULL
 ..@ inherit.blank: logi TRUE
$ point                          : <ggplot2::element_point>
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 ..@ shape        : num 19
 ..@ size         : num 1.5
 ..@ fill         : chr "white"
 ..@ stroke       : num 0.5
 ..@ inherit.blank: logi TRUE
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$ geom                           : <ggplot2::element_geom>
 ..@ ink        : chr "black"
```

```
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 ..@ accent     : chr "#3366FF"
 ..@ linewidth  : num 0.5
 ..@ borderwidth: num 0.5
 ..@ linetype   : int 1
 ..@ bordertype : int 1
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 ..@ fontsize   : num 3.87
 ..@ pointsize  : num 1.5
 ..@ pointshape : num 19
 ..@ colour     : NULL
 ..@ fill       : NULL
$ spacing                         : 'simpleUnit' num 5.5points
 ..- attr(*, "unit")= int 8
$ margins                         : <ggplot2::margin> num [1:4] 5.5 5.5 5.5 5.5
$ aspect.ratio                    : NULL
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$ axis.title.x                    : <ggplot2::element_text>
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 ..@ face         : NULL
 ..@ italic       : chr NA
 ..@ fontweight   : num NA
 ..@ fontwidth    : num NA
 ..@ colour       : NULL
 ..@ size         : NULL
 ..@ hjust        : NULL
 ..@ vjust        : num 0
 ..@ angle        : NULL
```

```
 ..@ lineheight   : NULL
 ..@ margin       : <ggplot2::margin> num [1:4] 0 0 2.75 0
 ..@ debug        : NULL
 ..@ inherit.blank: logi TRUE
$ axis.title.x.bottom             : NULL
$ axis.title.y                    : <ggplot2::element_text>
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 ..@ face         : NULL
 ..@ italic       : chr NA
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 ..@ fontwidth    : num NA
 ..@ colour       : NULL
 ..@ size         : NULL
 ..@ hjust        : NULL
 ..@ vjust        : num 1
 ..@ angle        : num 90
 ..@ lineheight   : NULL
 ..@ margin       : <ggplot2::margin> num [1:4] 0 2.75 0 0
 ..@ debug        : NULL
 ..@ inherit.blank: logi TRUE
$ axis.title.y.left               : NULL
$ axis.title.y.right              : <ggplot2::element_text>
 ..@ family       : NULL
 ..@ face         : NULL
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 ..@ fontweight   : num NA
 ..@ fontwidth    : num NA
 ..@ colour       : NULL
 ..@ size         : NULL
 ..@ hjust        : NULL
 ..@ vjust        : num 1
 ..@ angle        : num -90
 ..@ lineheight   : NULL
 ..@ margin       : <ggplot2::margin> num [1:4] 0 0 0 2.75
 ..@ debug        : NULL
 ..@ inherit.blank: logi TRUE
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 ..@ face         : NULL
 ..@ italic       : chr NA
 ..@ fontweight   : num NA
 ..@ fontwidth    : num NA
 ..@ colour       : chr "#4D4D4DFF"
```

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 ..@ fontweight   : num NA
 ..@ fontwidth    : num NA
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 ..@ margin       : <ggplot2::margin> num [1:4] 2.2 0 0 0
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 ..@ angle        : NULL
 ..@ lineheight   : NULL
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 ..@ debug        : NULL
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$ axis.text.x.bottom              : NULL
$ axis.text.y                     : <ggplot2::element_text>
 ..@ family       : NULL
 ..@ face         : NULL
 ..@ italic       : chr NA
```

```
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 ..@ colour       : NULL
 ..@ size         : NULL
 ..@ hjust        : num 1
 ..@ vjust        : NULL
 ..@ angle        : NULL
 ..@ lineheight   : NULL
 ..@ margin       : <ggplot2::margin> num [1:4] 0 2.2 0 0
 ..@ debug        : NULL
 ..@ inherit.blank: logi TRUE
$ axis.text.y.left                : NULL
$ axis.text.y.right               : <ggplot2::element_text>
 ..@ family       : NULL
 ..@ face         : NULL
 ..@ italic       : chr NA
 ..@ fontweight   : num NA
 ..@ fontwidth    : num NA
 ..@ colour       : NULL
 ..@ size         : NULL
 ..@ hjust        : num 0
 ..@ vjust        : NULL
 ..@ angle        : NULL
 ..@ lineheight   : NULL
 ..@ margin       : <ggplot2::margin> num [1:4] 0 0 0 2.2
 ..@ debug        : NULL
 ..@ inherit.blank: logi TRUE
$ axis.text.theta                 : NULL
$ axis.text.r                     : <ggplot2::element_text>
 ..@ family       : NULL
 ..@ face         : NULL
 ..@ italic       : chr NA
 ..@ fontweight   : num NA
 ..@ fontwidth    : num NA
 ..@ colour       : NULL
 ..@ size         : NULL
 ..@ hjust        : num 0.5
 ..@ vjust        : NULL
 ..@ angle        : NULL
 ..@ lineheight   : NULL
 ..@ margin       : <ggplot2::margin> num [1:4] 0 2.2 0 2.2
 ..@ debug        : NULL
 ..@ inherit.blank: logi TRUE
```

```
$ axis.ticks                      : <ggplot2::element_line>
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 ..@ linetype     : NULL
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 ..@ linejoin     : NULL
 ..@ arrow        : logi FALSE
 ..@ arrow.fill   : chr "#333333FF"
 ..@ inherit.blank: logi TRUE
$ axis.ticks.x                    : NULL
$ axis.ticks.x.top                : NULL
$ axis.ticks.x.bottom             : NULL
$ axis.ticks.y                    : NULL
$ axis.ticks.y.left               : NULL
$ axis.ticks.y.right              : NULL
$ axis.ticks.theta                : NULL
$ axis.ticks.r                    : NULL
$ axis.minor.ticks.x.top          : NULL
$ axis.minor.ticks.x.bottom       : NULL
$ axis.minor.ticks.y.left         : NULL
$ axis.minor.ticks.y.right        : NULL
$ axis.minor.ticks.theta          : NULL
$ axis.minor.ticks.r              : NULL
$ axis.ticks.length               : 'rel' num 0.5
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$ axis.ticks.length.x.bottom      : NULL
$ axis.ticks.length.y             : NULL
$ axis.ticks.length.y.left        : NULL
$ axis.ticks.length.y.right       : NULL
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$ axis.minor.ticks.length.y       : NULL
$ axis.minor.ticks.length.y.left  : NULL
$ axis.minor.ticks.length.y.right : NULL
$ axis.minor.ticks.length.theta   : NULL
$ axis.minor.ticks.length.r       : NULL
$ axis.line                       : <ggplot2::element_blank>
$ axis.line.x                     : NULL
```

```
$ axis.line.x.top                 : NULL
$ axis.line.x.bottom              : NULL
$ axis.line.y                     : NULL
$ axis.line.y.left                : NULL
$ axis.line.y.right               : NULL
$ axis.line.theta                 : NULL
$ axis.line.r                     : NULL
$ legend.background               : <ggplot2::element_rect>
 ..@ fill         : NULL
 ..@ colour       : logi NA
 ..@ linewidth    : NULL
 ..@ linetype     : NULL
 ..@ linejoin     : NULL
 ..@ inherit.blank: logi TRUE
$ legend.margin                   : NULL
$ legend.spacing                  : 'rel' num 2
$ legend.spacing.x                : NULL
$ legend.spacing.y                : NULL
$ legend.key                      : NULL
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 ..- attr(*, "unit")= int 3
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$ legend.key.width                : NULL
$ legend.key.spacing              : NULL
$ legend.key.spacing.x            : NULL
$ legend.key.spacing.y            : NULL
$ legend.key.justification        : NULL
$ legend.frame                    : NULL
$ legend.ticks                    : NULL
$ legend.ticks.length             : 'rel' num 0.2
$ legend.axis.line                : NULL
$ legend.text                     : <ggplot2::element_text>
 ..@ family       : NULL
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 ..@ colour       : NULL
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 ..@ hjust        : NULL
 ..@ vjust        : NULL
 ..@ angle        : NULL
 ..@ lineheight   : NULL
```

```
  ..@ margin       : NULL
  ..@ debug        : NULL
  ..@ inherit.blank: logi TRUE
 $ legend.text.position            : NULL
 $ legend.title                    : <ggplot2::element_text>
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  ..@ face         : NULL
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  ..@ colour       : NULL
  ..@ size         : NULL
  ..@ hjust        : num 0
  ..@ vjust        : NULL
  ..@ angle        : NULL
  ..@ lineheight   : NULL
  ..@ margin       : NULL
  ..@ debug        : NULL
  ..@ inherit.blank: logi TRUE
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 $ legend.position.inside          : NULL
 $ legend.direction                : NULL
 $ legend.byrow                    : NULL
 $ legend.justification            : chr "center"
 $ legend.justification.top        : NULL
 $ legend.justification.bottom     : NULL
 $ legend.justification.left       : NULL
 $ legend.justification.right      : NULL
 $ legend.justification.inside     : NULL
  [list output truncated]
 @ complete: logi TRUE
 @ validate: logi TRUE
```

Gene Expression Plot

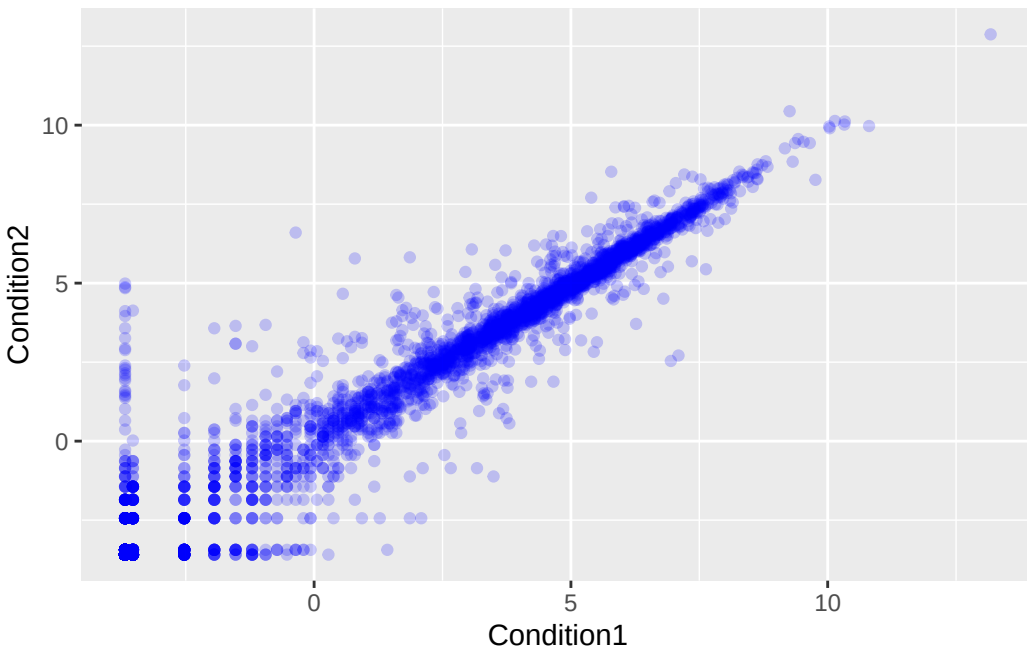
Read some data on the effects of GLP-1 inhibitor (drug) on gene expression values

```r
url <- "https://bioboot.github.io/bimm143_S20/class-material/up_down_expression.txt"
genes <- read.delim(url)
head(genes)
```

```
        Gene Condition1 Condition2      State
1      A4GNT -3.6808610 -3.4401355 unchanging
2       AAAS  4.5479580  4.3864126 unchanging
3      AASDH  3.7190695  3.4787276 unchanging
4       AATF  5.0784720  5.0151916 unchanging
5       AATK  0.4711421  0.5598642 unchanging
6 AB015752.4 -3.6808610 -3.5921390 unchanging
```

Version 1 plot - start simple by getting some ink on the page.

```
ggplot(genes) +
  aes(x=Condition1, y=Condition2) +
  geom_point(col= "blue", alpha=0.2)
```

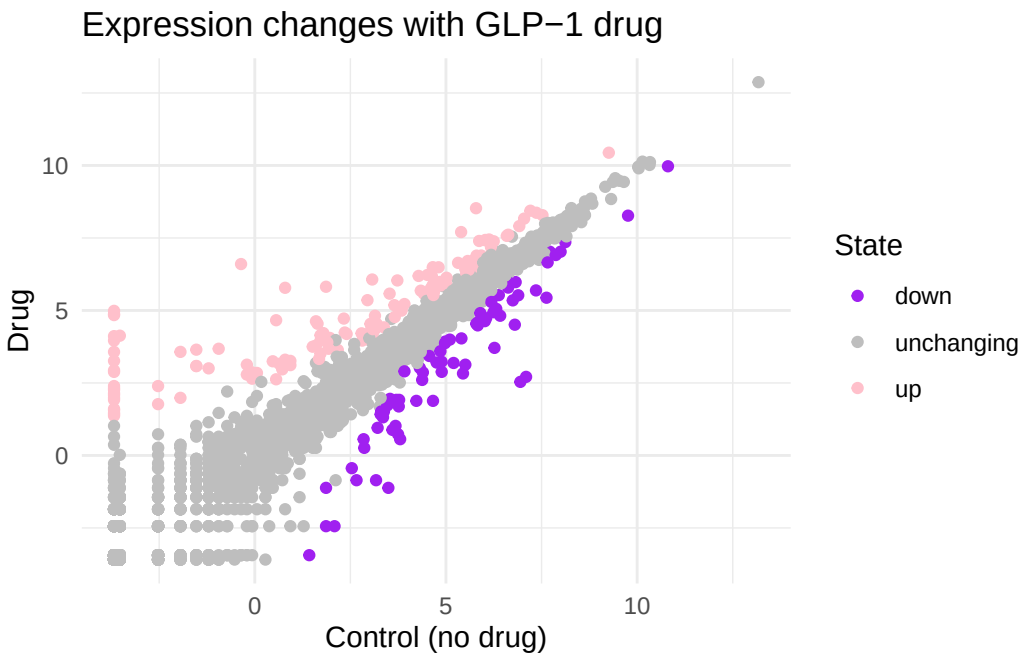


Let's color by `State` up, down, or no change.

```
table( genes$State )
```

```
      down unchanging         up
        72       4997        127
```

```r
ggplot(genes) +
  aes(x=Condition1, y=Condition2, col=State) +
  geom_point() +
  scale_color_manual(values =c("purple","gray","pink")) +
  labs(x ="Control (no drug)", y="Drug", title ="Expression changes with GLP-1 drug") +
  theme_minimal()
```



Going further with gapminder

Here we explore the famous `gapminder` dataset with some custom plots.

```r
# File location online
url <- "https://raw.githubusercontent.com/jennybc/gapminder/master/inst/extdata/gapminder.tsv

gapminder <- read.delim(url)
head(gapminder)
```

```
      country continent year lifeExp      pop gdpPercap
1 Afghanistan      Asia 1952  28.801  8425333  779.4453
2 Afghanistan      Asia 1957  30.332  9240934  820.8530
3 Afghanistan      Asia 1962  31.997 10267083  853.1007
```

```
4 Afghanistan      Asia 1967  34.020 11537966  836.1971
5 Afghanistan      Asia 1972  36.088 13079460  739.9811
6 Afghanistan      Asia 1977  38.438 14880372  786.1134
```

Q. How many rows does this datset have?

```
nrow(gapminder)
```

```
[1] 1704
```

How many different continents are in this dataset?

```
table(gapminder$continent)
```

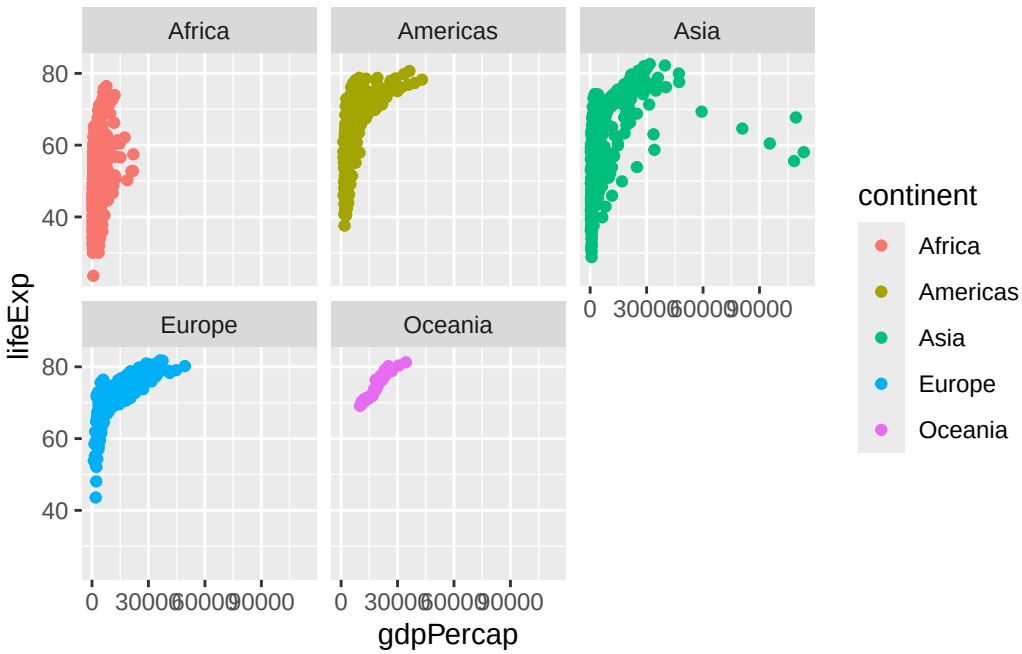
```
  Africa Americas     Asia   Europe  Oceania 
     624      300      396      360       24 
```

Version 1 plot gdpPercap vs LifeExp for all rows

```
ggplot(gapminder) +
  aes(gdpPercap, lifeExp, col=continent) +
geom_point()
```

I want to see a plot for each continent - in ggplot lingo this is called "faceting"

```
ggplot(gapminder) +
  aes(gdpPercap, lifeExp, col=continent) +
geom_point() +
facet_wrap(~continent)
```



First look at the dplyr package

Another add-on package with a function called `filter()` that we want to use

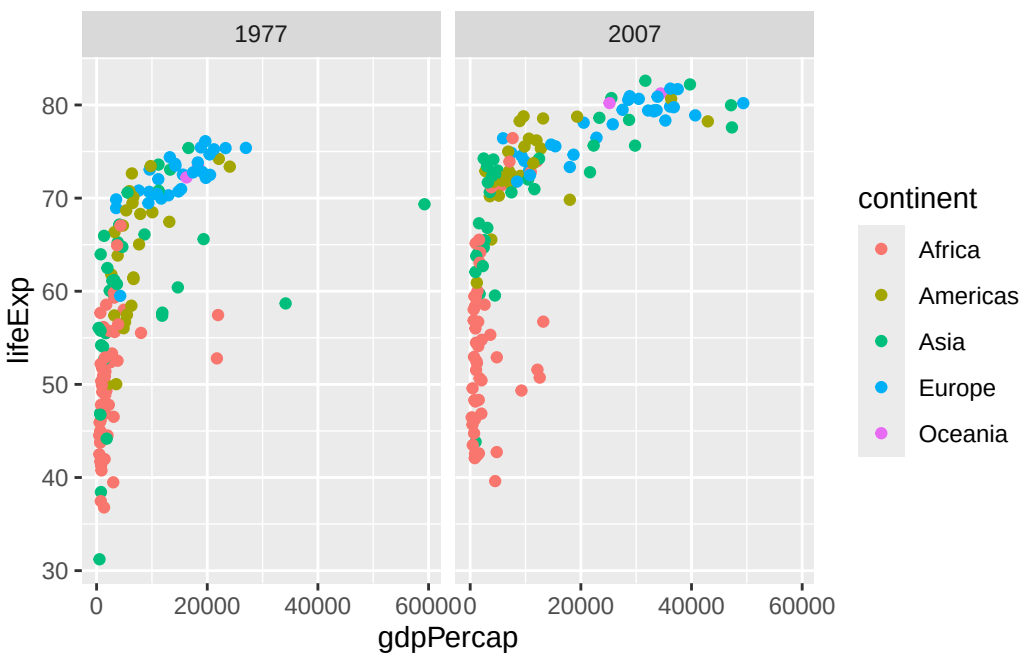
```
library(dplyr)
```

```
Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

    filter, lag
```

```
The following objects are masked from 'package:base':

    intersect, setdiff, setequal, union
```

```r
filter(gapminder, year == 2007, country=="Ireland")
```

```
  country continent year lifeExp     pop gdpPercap
1 Ireland    Europe 2007  78.885 4109086     40676
```

```r
input <- filter(gapminder, year == 2007 | year == 1977)
```

```r
ggplot(input) +
aes(gdpPercap, lifeExp, col=continent) +
  geom_point() +
  facet_wrap(~year)
```